

PROJECT WRITE-UP

AI Policy Index

A unified open-source dashboard mapping AI governance, readiness scores, and public opinion across 193 countries — built with a full data pipeline and an interactive choropleth visualization.

Role: Solo — data engineering + front-end **Stack:** Python · Next.js · TypeScript

Year: 2025–2026

OVERVIEW

Global AI policy is moving fast, but the data about it is scattered across academic PDFs, government reports, and paywalled index providers. The AI Policy Index consolidates fourteen sources into one queryable dataset — and surfaces it as a live, interactive world map.

The project spans two distinct layers: a Python data pipeline that fetches, normalizes, and validates heterogeneous source data; and a Next.js web application that renders 195 countries as a color-coded choropleth, updated with a year slider and switchable indicators.

It was built as a public-interest tool for researchers, journalists, and policy teams who want to understand the global AI governance

landscape at a glance — and drill down into a country's position across multiple dimensions without stitching together spreadsheets.

195

COUNTRIES MAPPED

14

DATA SOURCES

18

INDICATORS

2023-25

YEARS COVERED

THE PROBLEM

AI governance data is fragmented by design

Every major AI policy dataset lives in its own silo. The Oxford Insights Government AI Readiness Index is published as a pay-to-access PDF with CSV scores behind a form gate. Stanford HAI's AI Index ships as a 300-page annual report. Ipsos opinion data comes as formatted press releases. OECD policy records require scraping a JavaScript-heavy portal.

This fragmentation imposes a large upfront cost on anyone who wants to ask comparative questions: Which regions have both high AI readiness and low public trust? Which countries passed the most AI laws between 2020 and 2025? How does public excitement about AI correlate with government investment?

No existing tool answers these questions in one place. The AI Policy Index was built to fill that gap — first as a structured dataset, then as a visualization layer on top.

"The goal was not just to display data, but to make data joinable — keyed on ISO 3166-1 alpha-3 country codes and year, so any source can be compared against any other with a single merge."

DATA SOURCES

Three categories of evidence, integrated into one schema

Sources were selected for geographic breadth, methodological transparency, and public availability. They fall into three functional categories:

READINESS

Oxford Insights Government AI Readiness Index

Annual composite score covering government strategy, technology sector capacity, and data infrastructure. Available for 193–195 countries, 2023–2025. The 2025 edition restructured into 6 pillars; overall scores were re-derived via dimension-count weighted averaging (mean absolute error vs. stated values: ~1.4 points).

OPINION

Ipsos AI Monitor 2024

Public opinion survey across 32 countries. Four indicators extracted: nervousness about AI, excitement about AI, expectation that AI will change daily life, and trust in corporate data protection under AI. Coverage skews toward high- and upper-middle-income countries.

INVESTMENT

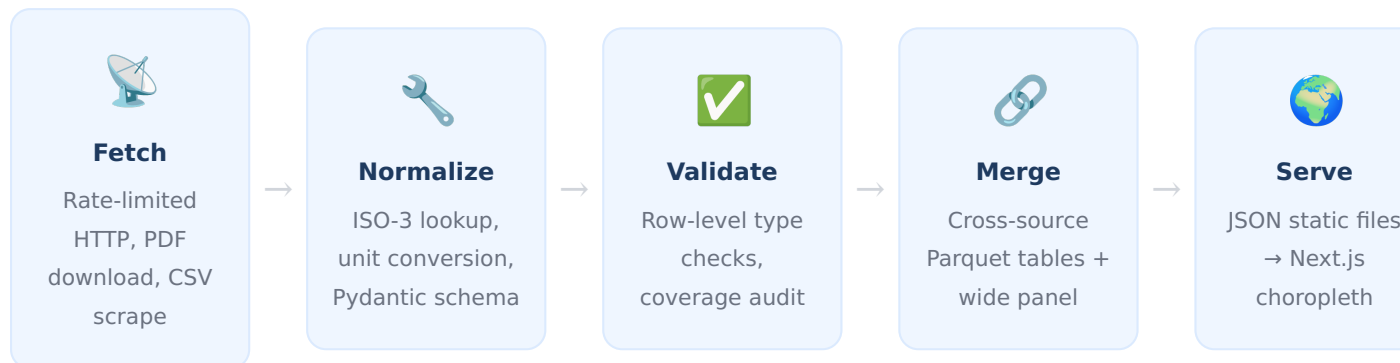
Stanford HAI 2026 AI Index Report

Country-level economic and adoption data for 2025: generative AI adoption rate (30 countries), private AI investment in USD billions (15 countries), AI job postings as share of all job postings (22 countries), and cumulative AI laws passed 2016–2025 across G20 members (14 countries).

The pipeline is designed to ingest ten additional sources — including the OECD AI Policy Navigator, Global Index on Responsible AI (GIRAI), World Risk Poll, and Pew Research surveys — as data access permits. Each source module follows the same interface: fetch, normalize to a shared Pydantic schema, validate ISO-3 country codes, and emit a per-source Parquet file.

ARCHITECTURE

A two-tier architecture: pipeline then presentation



Back-end (Python / uv). Each data source lives in its own module under `src/sources/`. Modules are discovered automatically via a source registry in `config/sources.yaml` and run through a shared CLI: `uv run python -m src.cli run-all`. Intermediate data is cached as per-source Parquet files in `data/interim/`; the merge step then joins them into three normalized output tables — indicators, policies, and `public_opinion` — plus a wide-format country-year panel for regression-ready analysis.

Country normalization is handled centrally via ISO 3166-1 alpha-3. Every source uses different country name conventions ("Czechia" vs. "Czech Republic", "USA" vs. "United States"). The `common/countries.py` module resolves these through `pycountry` with a hand-curated disambiguation layer for ambiguous cases (Kosovo, Taiwan, Western Sahara).

Front-end (Next.js 16 / React 19). The web application serves pre-built JSON files from `public/data/` — no server-side database query at render time. The map itself is an SVG rendered from Natural Earth TopoJSON (110m resolution), projected with D3's Equal Earth projection, and colored with a sequential scale computed dynamically from each indicator's value distribution.

Clicking any country routes to a per-country detail page. The year slider and indicator picker are controlled components that recompute the color map on every selection change without re-fetching data.



DESIGN DECISIONS

Key trade-offs and why they were made

DECISION	ALTERNATIVE CONSIDERED	RATIONALE
ISO-3 as the universal join key	Country name string matching	Names vary too much across sources (English vs. French, full vs. abbreviated). ISO-3 codes are stable and machine-verifiable; ambiguous names are caught at normalization time, not silently dropped.
Static JSON files served from public/	Database-backed API routes	Data updates monthly at most; a static approach eliminates infrastructure cost and latency, keeps the deployment a simple Vercel project, and makes the dataset directly downloadable.
Equal Earth SVG projection	Mercator, Robinson	Equal Earth preserves area ratios, avoiding the visual distortion that

DECISION	ALTERNATIVE CONSIDERED	RATIONALE
		makes high-latitude countries look disproportionately large — relevant when color-mapping a global policy indicator where small equatorial countries matter as much as large arctic ones.
Sequential D3 color scale computed from live values	Fixed min/max per indicator	Fixed scales would make poorly-covered indicators look artificially compressed. A distribution-aware scale (using the actual value spread in the current indicator/year slice) maximizes color contrast across the data that actually exists.
Per-source Parquet intermediates	Direct merge from raw files	Caching each source's normalized output lets individual sources be re-fetched and re-validated in isolation without re-running the entire pipeline — essential when one source's upstream changes or breaks.

PRODUCT FEATURES

What the interface delivers



Interactive choropleth map

195 countries colored by the selected indicator, with hover tooltips showing exact values and source attribution. Countries with no data are shown in a neutral gray with an explicit legend entry. Accessible via keyboard navigation and ARIA labels.



Year slider with gap awareness

A slider exposes all years for which any indicator has coverage. If the selected indicator has no data for the chosen year, the map blanks gracefully with an amber banner rather than silently displaying stale data. The default year is always the most recent with actual data.



18 switchable indicators

Grouped by source, spanning government AI readiness (overall + 6 pillars), public opinion (nervousness, excitement, life impact, trust), and economic indicators (investment, job postings, adoption rate, AI laws). Each carries metadata — source, methodology URL, description, coverage count — surfaced inline.



Country drill-down pages

Clicking any country routes to a dedicated page at `/country/[iso3]`, designed to show all available indicators for that country across years in one view.



Methodology transparency

A dedicated methodology page documents each data source, the approximation used for the 2025 Oxford overall score, known limitations (coverage gaps, survey sampling bias), and links to upstream reports. Disputed territories are acknowledged explicitly.

CHALLENGES

What was hard, and how it was resolved

The 2025 Oxford Insights restructure. Oxford changed its methodology between 2024 and 2025, replacing three pillars with six. The 2025 report publishes pillar scores but not a directly comparable overall score by the same formula as prior years. To maintain a consistent time series, overall scores were approximated by weighting the six pillars by their sub-dimension counts — a

defensible but imperfect reconstruction with a mean absolute error of ~ 1.4 points against values stated in the report. This limitation is documented on the methodology page.

Stanford HAI data extraction. The 2026 AI Index report publishes country-level data as narrative charts without machine-readable downloads. Values for generative AI adoption, private investment, AI job postings, and cumulative AI laws were extracted by reading the report directly and hardcoded as structured records in the pipeline — a tedious but reproducible approach that documents exactly which figure in which chapter each value came from.

React 19 peer dependency conflicts. The `@visx/geo` visualization library declares a peer dependency on React 16–18; React 19 was released after its last update. Resolving this required a project-level `.npmrc` with `legacy-peer-deps=true` for Vercel builds — not ideal, but less disruptive than forking or replacing the library while the upstream adds React 19 support.

Vercel deployment routing. Setting the Vercel root directory to `web/` did not automatically trigger Next.js framework detection; the platform treated the project as a static site, producing a 9-second build (vs. the 30–60 seconds a real Next.js compile requires) and a 404 on all routes. The fix was adding a `vercel.json` declaring `"framework": "nextjs"`, which forces correct framework detection regardless of dashboard settings.

Where the project is heading

The data pipeline already has stub modules for ten additional sources — OECD AI Policy Navigator, UNESCO Readiness Assessment Methodology, the Global Index on Responsible AI, Eurobarometer, World Risk Poll, and others — waiting on data access or JavaScript-rendering workarounds. As these come online, coverage will deepen from the current 32-country opinion layer to 100+ countries.

The front-end roadmap includes the country comparison view (side-by-side radar charts across indicators), a full country profile page with time-series sparklines, and a downloadable CSV export for researchers who want to use the merged dataset directly.

A GitHub Actions workflow is configured for monthly refresh on the 30th of each month, keeping the live site current as Oxford Insights, Stanford HAI, and Ipsos publish new editions.

AI Policy Index — built by Anna Siamionava

[Live site](#) · [GitHub repository](#) · annase2004@gmail.com

Data sources: Oxford Insights · Ipsos AI Monitor · Stanford HAI AI Index